

Process for no good reason?

1. Please consult the Activity and Milestone graph. What are all the possible paths between milestones to completion?

Ans:

The possible paths are:

1. 10 → 30 → 40 → 50 → 80 (activities A, B, C, D)
2. 10 → 30 → 50 → 80 (activities A, E, D)
3. 10 → 30 → 60 → 50 → 80 (activities A, F, G, D)
4. 10 → 20 → 60 → 50 → 80 (activities H, I, G, D)
5. 10 → 20 → 70 → 80 (activities H, J, K)

These routes represent all the ways work can logically flow from project start to final completion.

2. Please consult the Activity and Milestone graph. Describe the shortest start-finish relationship.

Ans:

The shortest start-to-finish path in the Activity and Milestones graph is:

10 → 30 → 50 → 80

This corresponds to activities A, E, and D in sequence. Activity A (10→30) takes 2 weeks, activity E (30→50) takes 2 weeks, and activity D (50→80) takes 2 weeks, giving a total of 6 weeks, which is shorter than any other complete path from project start to completion.

3. Please consult the Activity and Milestone graph. What is the duration of all critical paths? Include all milestones and times.

Ans:

The critical path is the longest start to finish route through the network, so any path with the maximum total duration is critical. In this graph, the duration of each complete path from milestone 10 to milestone 80 is:

- Path 1: 10 → 30 → 40 → 50 → 80, with activities A (2 weeks) + B (2 weeks) + C (2 weeks) + and D (2 weeks) = 8 weeks.
- Path 2: 10 → 30 → 60 → 50 → 80, with activities A (2 weeks), F (3 weeks), G (1 week) + D (2 weeks) = 8 weeks.
- Path 3: 10 → 20 → 70 → 80, with activities H (1 week) + J (3 weeks) + K (4 weeks) = 8 weeks.

Each of these three paths has the same longest duration of 8 weeks, so there are three critical paths in the network.

4. Please consult the Activity and Milestone graph. What is the duration of all subcritical paths? Include all milestones and times.

Ans:

Subcritical paths are the complete start to finish routes that are shorter than the 8 week critical paths. The paths are:

- Path 1: 10 → 30 → 50 → 80, with activities A (2weeks) + E (2weeks) + D (2weeks) = 6 weeks
- Path 2, 10 → 20 → 60 → 50 → 80, with activities H (1weeks) + I (3weeks) + G (1weeks) + D (2weeks) = 7 weeks

Hence there are two subcritical paths and total duration for these subcritical paths ranges from 6 to 7 weeks.

5. What are the consequences of removing the blue activities in the Removal of Processes figure? Can you still complete your project? Describe all viable paths (start-to-finish relationship) in terms of milestones.

Ans:

Removing the blue activities in the Removal of Processes graph cuts out several shorter and alternate routes through the network, but it does not disconnect the start milestone (10) from the finish milestone (80). The project can still be completed; however, there are fewer options to reach completion, and some of the faster paths disappear, which can increase overall project duration and reduce flexibility if delays occur on the remaining routes.

With only the red activities left, the viable start-to-finish paths are:

10 → 30 → 40 → 50 → 80

10 → 20 → 60 → 50 → 80

10 → 30 → 20 → 60 → 50 → 80

These are the paths that still provide a continuous forward sequence of milestones from 10 to 80 after all blue activities are removed.

Feedback

Feedback from grader

First one is wrong (mind the direction of the arrow), other viable paths are:

10 → 30 → 20 → 60 → 70 → 20 → 60 → 50 → 80

10 → 20 → 60 → 70 → 20 → 60 → 50 → 80

6. What might be the consequences of removing the blue activities in the Removal of Processes figure in terms of technical debt and scheduling?

Ans:

Removing the blue activities trims the process and can look attractive from a scheduling perspective, because there are fewer handoffs and less visible work to track. In the short term, the schedule may appear simpler and easier to manage, and the team can focus on the remaining red activities without worrying about the overhead of extra routes or checks. However, if those blue activities were providing useful validation, documentation, or refactoring steps, cutting them out can create technical debt: issues are pushed downstream instead of being resolved early, which often leads to rework, defects, and higher long-term maintenance effort. Over time, that debt can slow future changes and make the system harder to evolve, even if the initial schedule looked faster.

From a scheduling standpoint, removing alternate blue paths also reduces slack and flexibility in the network. With fewer ways to move from start to finish, more work is forced onto the remaining routes, which can make the effective critical path longer and more fragile. Any delay on one of the surviving activities has a higher chance of delaying the whole project because there are no “backup” paths to absorb problems or allow resequencing of work. So while eliminating blue activities may reduce immediate complexity, it can trade short-term schedule gains for higher technical risk and tighter, less resilient timelines later in the project.

7. What are the consequences of removing the blue activities in the Removal of Processes figure? How many of these viable paths involve revisiting the same milestone?

Ans:

Removing the blue activities removes the rework loops and side-branches that previously allowed the process to cycle back to earlier milestones. Without those blue steps, the network becomes a straightforward forward-moving structure: every remaining red activity moves from one milestone to the next without circling back. The project is still fully completable, but the workflow becomes more linear and less tolerant of mistakes, since none of the removed steps send the team backward for revisions.

When we look at the viable start-to-finish paths that still remain:

- 10 → 30 → 40 → 50 → 80
- 10 → 30 → 50 → 80

- $10 \rightarrow 30 \rightarrow 60 \rightarrow 50 \rightarrow 80$
- $10 \rightarrow 20 \rightarrow 60 \rightarrow 50 \rightarrow 80$
- $10 \rightarrow 20 \rightarrow 70 \rightarrow 80$

None of these paths revisit the same milestone once it has been passed. Every path proceeds forward through a unique sequence. This is a direct consequence of removing the blue activities, because those blue steps represented the only backward-facing arrows that allowed a milestone to be reached twice.

So after the blue activities are removed, 0 of the viable paths involve revisiting a milestone.

Feedback

Feedback from grader

Only one of the paths you mentioned are viable:

- $10 \rightarrow 20 \rightarrow 60 \rightarrow 50 \rightarrow 80$

When I said "remove the blue paths", I wanted you to consider that they did not exist at all.